



Contents lists available at ScienceDirect

Journal of Pharmaceutical Sciences

journal homepage: www.jpharmsci.org

Pharmaceutics, Drug Delivery and Pharmaceutical Technology

Rapid Room-Temperature Aerosol Dehydration Versus Spray Drying: A Novel Paradigm in Biopharmaceutical Drying Technologies

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ARTICLE INFO

Article history:

Received 22 August 2023

Revised 1 October 2023

Accepted 1 October 2023

Available online xxx

Keywords:

Biopharmaceutical processing

Aerosol-assisted drying

Particle engineering

Spray drying

Protein degradation

ABSTRACT

To ensure the high quality of biopharmaceutical products, it is imperative to implement specialized unit operations that effectively safeguard the structural integrity of large molecules. While lyophilization has long been a reliable process, spray drying has recently garnered attention for its particle engineering capabilities for the pulmonary route of administration. However, maintaining the integrity of biologics during spray drying remains a challenge. To address this issue, we explored a novel dehydration system based on aerosol-assisted room-temperature drying of biological formulations recently developed at Princeton University, called Rapid Room-Temperature Aerosol Dehydration. We compared the quality attributes of the bulk powder of biopharmaceutical products manufactured using this drying technology with that of traditional spray drying. For all the fragment antigen-binding formulations tested, in terms of protein degradation and aerosol performance, we were able to achieve a better product quality using this drying technology compared to the spray drying technique. We also highlight areas for improvement in future prototypes and prospective commercial versions of the system. Overall, the offered dehydration system holds potential for improving the quality and diversity of biopharmaceutical products and may pave the way for more efficient and effective production methods in the biopharma industry.

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Introduction

The market for biopharmaceuticals has experienced significant growth, with a value of approximately \$400 billion in 2021, representing an increase of nearly 50 % from 2016. By 2027 the market is projected to generate a revenue of \$534 billion.¹ Today, modern best-seller drugs are more likely to be complex biologics than small molecules, which accounted for 80 % of the most sold pharmaceuticals at the turn of the millennium.¹ For the first time ever, there are now roughly as many biologic blockbusters as there are small molecule bestsellers.¹ Though most biologics are typically delivered via injections or infusions due to low oral bioavailability, inhalation of dry powder is being explored because of its potential benefits. These benefits could include improved shelf life, better patient compliance and comfort, controlled drug release capabilities, and possibly enhanced efficacy in certain conditions, although the effectiveness varies and needs further investigation.² Therefore, several attempts have been made to commercialize inhalation products for biologics using spray

drying since 2006, beginning when Exubera was approved as the first insulin inhaled powder for type 1 and 2 diabetes treatment. The spray drying technology has been successfully utilized to dry a wide variety of heat-labile compounds, proteins and cell products over the last decade. Notable examples include probiotic cells, nitrogen-fixing bacteria,^{3,4} insulin,⁵ bovine serum albumin,⁶ plasma coagulation proteins fibrinogen⁷ and thrombin,⁸ and vaccines.⁹ In addition to producing inhalable powder, spray drying offers numerous advantages over other drying technologies, including energy sustainability, being a continuous process, and the capability to produce sustained drug delivery systems.¹⁰

Although spray drying presents certain advantages, it's important to note that the stress it induces can substantially affect protein stability.¹¹ The process of spray drying involves atomizing a liquid protein solution into small droplets, which are then rapidly dried to form a powder. These processes can subject the protein to various stresses, including thermal stress, mechanical stress, and adsorption at the air-liquid interface.^{11,12} These stresses can disrupt the protein's native structure, exposing its hydrophobic regions. Such exposure can subsequently lead to protein denaturation, aggregation, and a resulting loss of activity.¹³ A recent study using a high-resolution

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hydrogen-deuterium exchange mass spectrometry (HDX-MS) on insoluble aggregates in reconstituted spray dried powder demonstrated that the antigen binding site was involved in the aggregate formation.¹⁴ The results suggest that the stresses from spray drying disrupt the protein structure, exposing the hydrophobic residues in heavy-chain complementarity-determining region 1 (CDR-1). Subsequently, insoluble aggregates are formed from these disrupted proteins through hydrophobic interactions upon reconstitution with water.¹⁴ The extent of protein stability loss during spray drying can vary depending on the specific process equipment and conditions used and the type of protein being dried.^{13,15–17} It has been suggested that, compared to the smaller scales, the commercial spray drying process causes more protein denaturation, exposing more hydrophobic amino acids and increasing protein aggregation.¹⁸ However, the use of excipients during spray drying can help stabilize proteins by aiding in interfacial stabilization, improving yield, and enhancing processability of the spray-dried powders.^{11,13,19} Overall, spray drying stresses can have a detrimental effect on protein stability, but the specific impact can vary depending on the process conditions and the use of stabilizing excipients.^{19,20}

Therefore, there are still technical setbacks and challenges that need to be overcome. Degradation of protein (in both soluble and insoluble aggregates) and loss of bioactivity during spray drying are the two main issues,²¹ at least during bulk powder manufacturing step. Unfolding of a protein's three-dimensional structure may lead to the formation of noncovalent aggregates. This can result in irreversible damage to the proteins, thereby reducing their bioactivity

and efficacy. Moreover, it can heighten the risk of immunogenicity.²¹ Among other spray drying stresses, atomization and drying imposes large stresses on proteins, and can cause soluble and insoluble aggregates.²² Using less stressful alternative atomizing techniques and/or lowering manufacturing temperature could mitigate the observed challenges. To that end, this work builds on extensive prior development studies done at Princeton University^{23–25} and deploys a novel drying technique that mitigates two common sources of protein degradation associated with spray drying of biologics – namely, high temperatures and high gas-liquid interfacial stresses. The new technique is based on gentle atomization and room temperature drying, which involves two steps: atomizing the precursor solution into an aerosol of fine droplets using jet-driven liquid film disintegration,²⁴ and room temperature dehydration²⁵ (Fig. 1(b)). The resulting powders are collected using an electrostatic collector or a filter. Despite being a different process, this dehydration technique still offers some of the main advantages of spray drying, such as semi-continuous processing, particle design, and short manufacturing time. A schematic of each of the two deployed instruments and their mechanisms of action is provided in Fig. 1.

In this study we compare the performance of the two dehydration methods for biologic synthesis and report that the new technique has several advantages over spray drying. Overall, the results of the study highlight the potential of the new Rapid Room-Temperature Aerosol Drying (RTAD) technique²⁵ as a viable alternative to traditional spray drying. The approach offers a gentle and efficient way to produce powders with unique characteristics, making it a promising method

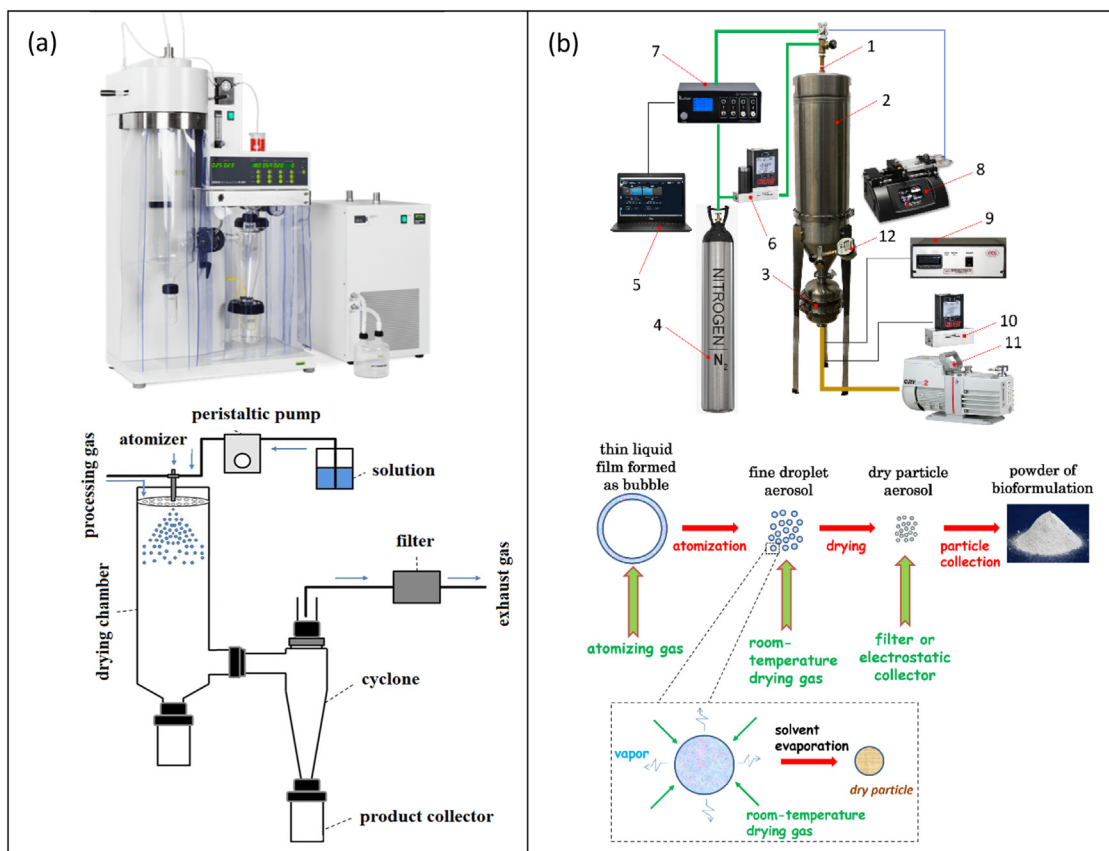


Figure 1. Drying technologies used in this study. (a) Photograph and schematic of Buchi Mini Spray Dryer B-290 (schematic is adopted from the B-290 manual). (b) Photograph and schematic of the laboratory-scale prototype of Rapid Room-Temperature Aerosol Drying (RTAD) system (schematic diagram of RTAD is adopted from²⁵). Here, 1 – aerosol generator assembly, 2 – drying chamber, 3 – filter assembly for collecting dry product, 4 – nitrogen cylinder to supply drying gas, 5 – computer, 6 – mass flow controller of drying gas flow, 7 – microfluidic flow controller, 8 – syringe pump to supply liquid biopharmaceutical formulation, 9 – hygrometer, 10 – flow meter, 11 – gas pump, 12 – manometer.

for biopharmaceutical applications. However, there are areas for improvement, specifically regarding production yield and residual moisture content that need to be addressed in future works.

Materials and Methods

Materials

A FAB (fragment antigen-binding) was formulated in two buffers and with two strengths (see Table 1). After feedstock dehydration processes, the powders were dispensed into pre-sterilized glass vials to do the quality tests outlined in the following section. The intent of using different buffers was to show the potential of the new system for synthesis of wider formulations. For each formulation, 60 mL of bulk feedstock was prepared, split, and processed. It is worth noting that the team from Princeton University, who conducted the experiments using the RTAD setup, received four containers labeled A-D (refer to Table 1) from AstraZeneca. The team was not privy to the specific details of the biopharmaceutical formulations.

Drying Systems

A Buchi Mini Spray Dryer B-290 (Buchi, Switzerland) and a novel RTAD drying technique were used to process the feedstock into powders. The experiments with spray dryer B-290 were conducted at AstraZeneca (Gaithersburg, MD, USA), and the experiments with RTAD were performed at the laboratory of the Complex Fluids Group at Princeton University (Princeton, NJ, USA). An open loop system with nitrogen, N₂, as the drying gas was used for the spray drying process. The same feedstocks mentioned in Table 1 were used in the two drying processes. Although the drying mechanisms and their associated operation parameters differ between the two technologies, to avoid introducing new variables, similar formulation and process conditions were selected for each drying process across all units. The process conditions are summarized in Table 2.

After the drying process, the collected powders were dispensed into 20R SCHOTT Borosilicate glass vials with associated rubber stoppers and caps, obtained from West Pharmaceutical Services, Inc. (Exton, PA, USA). The vials were washed, dried, and depyrogenated before use. The vials of collected RTAD powder samples were shipped in a refrigerated container from Princeton University to AstraZeneca's Gaithersburg facility. All the RTAD powder samples were analyzed and compared with spray dried powder samples at AstraZeneca. Further aliquoting for analytical testing were done in a glove box with N₂ flow and relative humidity RH < 2 % until assayed. For reconstitution medium and dilution, USP Sterile Water for Injection and Phosphate Buffered Saline (PBS) 1X were procured from Fisher Scientific (Waltham, MA, USA).

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Characterization Methods

Karl Fischer Titration

Residual moisture was tested using the Karl Fischer (KF) titration method (Mettler-Toledo OH, USA). Each reported value (%; w/w) is the mean of three determinations. This method involves dissolving a small quantity of the powder in a solvent, methanol, followed by the addition of a standardized Karl Fischer reagent solution, which reacts with any water in the sample to produce iodine ions. These ions are then titrated with the Karl Fischer reagent until the endpoint is reached. The volume of titrant solution needed to reach this endpoint is directly proportional to the quantity of water in the sample, thus enabling the determination of moisture content.

Micro-flow Imaging (MFI)

The MFI 5200 micro-flow imaging system (Protein Simple, USA) was used to measure particle count/concentration and capture images of particles. The sample cell was rinsed with water for injection from a freshly opened bottle with a typical particle count < 25 count/mL. 1 mL of each sample was loaded into the instrument using a micropipette tip with a barrier to minimize extrinsic particle burden. The samples were analyzed at room temperature, and MFI image analysis software (version 1.0.5864) was used to process and analyze collected images, with edge particles excluded from analyses. Particle size was measured as the diameter of a circle with an equivalent projected area, and particles stuck to the flow cell were counted only once. Particle size distributions were reported into several size categories, >1 μm, > 2 μm, > 5 μm, > 10 μm, and > 25 μm.

High Performance Size-exclusion Liquid Chromatography (HP-SEC) for Soluble Aggregates

HP-SEC was used to monitor changes in soluble protein levels within the reconstituted solutions. The solutions were first diluted with PBS and then 250 μg of sample protein solutions at the 10 mg/mL concentration were injected. Also, 100 mL of the supernatants were injected on an Agilent 1100 series system (Santa Clara, CA) HPLC system equipped with a TSKgel guard column and a TSKgel GS3000SWXL column (TOSOH Biosciences, Montgomeryville, PA) with a mobile phase containing 100 mM sodium phosphate dibasic and 0.05 % w/v sodium azide (MP Biomedicals, California), pH=6.7 at a flowrate of 0.6 mL/min. For each chromatogram, the Ultraviolet (UV) absorbance of the eluent measured at 280 nm was integrated to provide the total peak area. The percent soluble aggregates present after application of stress were calculated as the ratio of the areas under the peaks eluting prior to the monomer peak to the total peak area.

UV Absorption Spectroscopy at 280 nm Wavelength for Insoluble Aggregates

Quantification of insoluble aggregates was based on the UV absorbance of protein at 280 nm (A280) against a buffer blank on a Lunatic microfluidic device (Unchained Labs, MA, USA). The system can be used to measure insoluble protein aggregates by a filtration-based approach. Each protein sample was first filtered through a 0.22 μm membrane, which separates the soluble proteins from the insoluble aggregates.^{22,26} The insoluble aggregates were retained on the filter, while the soluble proteins passed through and were collected in a separate vial. Both the reconstituted solution containing the insoluble aggregates and the filtrate containing the soluble proteins were then scanned using the Lunatic UV system (Unchained Labs, Boston, MA) by injecting 2 μL of solution into micro-wells. The absorbance at

Table 1
Feedstocks used for the study.

	High strength	Low strength
Buffer I	A	B
Buffer II	D	C

Table 2
Processing conditions for spray drying and the Rapid Room-Temperature Aerosol Drying (RTAD) laboratory scale system.

Unit operation	Inlet temperature	Drying gas flowrate	Atomization gas flowrate	Liquid flowrate
Spray Drying	135°C	35 kg/h	0.3 kg/h	0.18 kg/h
RTAD	22°C	5 kg/h	0.45 kg/h	0.10 kg/h

280 nm was used to quantify the amount of protein present in each sample, and the difference in absorbance between the two samples, the reconstituted solution and the filtrate, was used to determine the amount of insoluble aggregates.

Particle Morphology and Size Analysis

A TM4000 (Hitachi, Japan) scanning electron microscope (SEM) operating at 15 kV was used for imaging and analysis of generated powder for both the drying techniques. To obtain SEM images of particles, the samples were first dispersed on carbon-coated grids and then coated with a thin 5 nm layer of gold. The coated samples were then imaged. The obtained images were then analyzed via an in-house image processing technique to mark particles and quantify size distribution. This code processes SEM images of particles by applying various image processing/computer vision techniques to detect and measure the size of particles in the image. The code imports the necessary image analysis libraries and loads the SEM image. Then it converts the image to grayscale, applies histogram equalization, and performs high-pass filtering and edge enhancement. It applies adaptive thresholding and morphological operations to extract particle contours and then filters out non-enclosed shapes and internal contours. The code then measures the size of each particle and calculates the mean particle diameter. Finally, it generates histograms and cumulative distribution curves of particle sizes and displays the results. A snippet of the code used in the study can be found in (<https://github.com/sadeghhjdj/SEM-Particle-size-git>).²⁷

Results and Discussion

We observed that the selection of a drying system may have a significant impact on the formation of soluble and insoluble aggregates in biopharmaceutical formulations.²⁸ Here we show that the deployment of the spray drying and RTAD systems can result in varying degrees of protein aggregation, which can potentially affect the stability, efficacy, and safety of the final drug product.

For the outlined formulations, and process conditions listed in Tables 1 and 2, respectively, we used HP-SEC and UV-vis methods to measure soluble and insoluble aggregates and reported the data in Fig. 2. According to this figure, soluble aggregates measured by HP-SEC do not show any difference between the two employed techniques, whereas insoluble aggregate percentages are lower (a 10-fold decrease) for RTAD compared to the spray dried powders. This change could be due to either atomization or thermal stresses or a combination of both during spray drying process that are mitigated in RTAD.

Fig. 3 illustrates the results of the MFI analysis, including both the subvisible particle population distribution and a visual representation. This method is commonly used during both the development

and commercial phases to monitor sub-visible particles with diameters ranging from 1 to 100 μm .²⁹ Based on the MFI data presented in Fig. 3, panel (a), it can be observed that, across all formulations, the reconstituted powders generated with RTAD consistently exhibit much lower levels of sub-visible particles compared to their spray dried counterparts. Additionally, the images provided in panels (b) and (c), captured during the MFI run, show a significantly clearer window for the powder synthesized by RTAD, further supporting the trend observed in panel (a). The general trend of the MFI results for all formulations is consistent with the trend observed for the insoluble aggregates (Fig. 2).

An important solid state characterization test that can determine the stability and aerosol performance of the generated powder is the residual moisture content after bulk manufacture.³⁰ Fig. 4 displays the residual moisture content (expressed as w/w %) measured by KF for the powders synthesized through the two studied techniques. The average measured moisture content for spray dried powders was around 2 %, whereas it was approximately 5 % for RTAD powders. The results of the study demonstrate that the moisture content in RTAD powders is consistently higher compared to that in the corresponding spray dried powders for all the formulations. The observed difference in moisture content could be influenced by multiple factors, including the higher temperature used in the spray drying process, which could lead to faster evaporation of moisture during drying. Additionally, spray drying offers good control and sealing, which can help to minimize the exposure of the particles to moisture during the process. On the other hand, the residence time of the particles in each tool is also an important factor to consider, as it can affect the level of moisture absorption by the powders. Moreover, powder handling procedures, such as dispensing post-manufacture (which was done in humidity-controlled environment for spray dried powders and not performed for RTAD powders), may contribute to the observed discrepancies. Further investigation is needed to determine the relative impact of each of these factors on the observed difference in moisture content between the two powder synthesis techniques.

The SEM images presented in Fig. 5 showcase the powders generated by the two technologies, with two representative images displayed for each technology. Despite using different formulations (see Table 1), the powders exhibited no discernible differences for each of the considered techniques, and hence only two representative images were chosen for brevity. Upon examination of the images in sub-plot (a) and (b), the spray-dried particles possess a raisin-like, shriveled shape, and have thin walls, as evidenced by the broken pieces observed in the image. On the other hand, the particles generated by RTAD appear more spherical, with no broken pieces visible, and thicker walls. These findings are consistent with previous reports that particles produced by RTAD exhibit spherical morphologies with solid-core microstructure due to smaller initial droplet diameter and lower Peclet numbers.²⁵

The observed changes in morphology between the two drying systems may be linked to the observed protein degradation behavior. In the case of the spray-dried powder, we observed a brittle structure, likely resulting from the higher drying rates inherent in the spray drying process.³¹ These higher drying rates could lead to protein structural changes, further exacerbating protein degradation.

During the drying process, hydrogen bonds that originally formed with water in the aqueous solution are replaced by bonds with excipients in the solid state.³² In spray drying, the rapid drying process, along with sudden shifts in pH and osmolality, may potentially compromise the protection of proteins as the replacement of hydrogen bonds by excipients could be disrupted.³³

Conversely, in the RTAD process, the production of spherical particles results from a gentler moisture removal process. This gentler approach allows protective excipients to shield proteins against

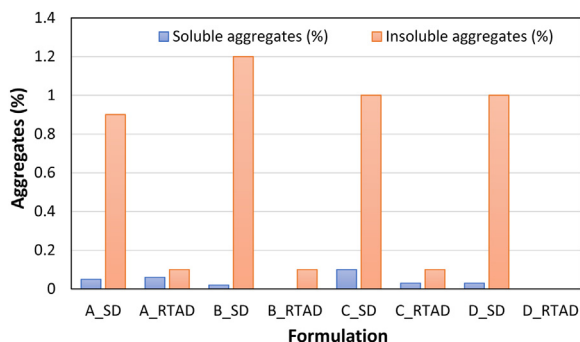


Figure 2. The impact of drying system on the production of soluble and insoluble protein aggregates across different formulations. Here A_SD to D_SD label the powders obtained by spray drying; A_RTAD to D_RTAD denote the powders obtained by the RTAD system.

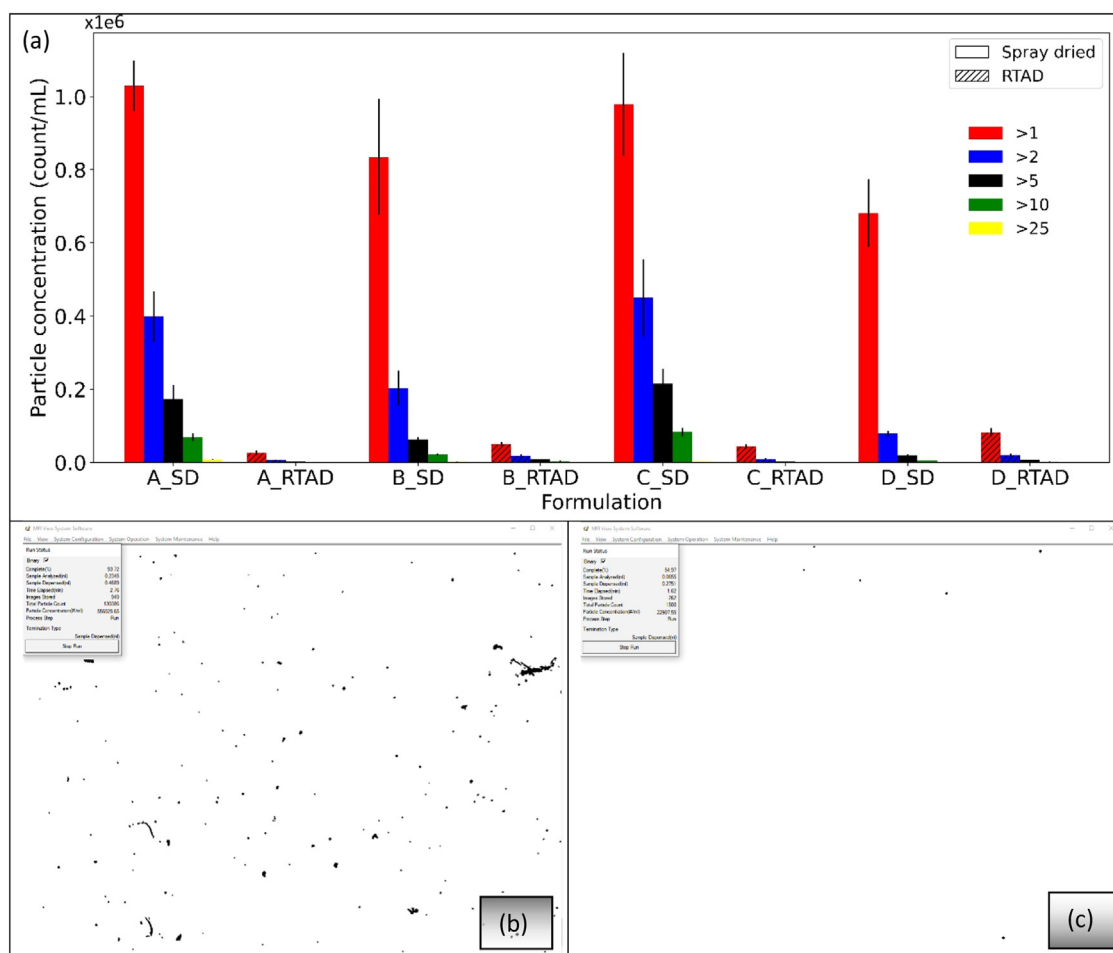


Figure 3. Concentration of sub-visible particles after powder reconstitution, as measured by MFI instrument for spray drying and RTAD systems across all the formulations. (a) Bar-chart of particle populations for various particle size categories 1–100 μm . (b), (c) Binary images from the computer screen of MFI instrument showing particle aggregates after reconstitution of the powders obtained, respectively from spray drying and RTAD systems.

sudden conformational changes, ultimately preserving the integrity of the protein structure.

To analyze the particle size distributions of the powders, a specialized image processing tool was adopted to process graphical information from the obtained SEM images. More information on the operation and capabilities of the developed image analysis tool is provided in the Appendix. Using this tool, size distribution data were assessed and averaged across various SEM images acquired for each formulation (with different microscope magnifications and at different locations on SEM powder sample) and across various formulations (see Table 1) used for each drying technology. The obtained representative number-weighted particle size distributions are illustrated in Fig. 6. The results presented in Fig. 6 indicate that the count mean particle diameter for the spray-dried powder was $D_{50} = 6.3 \mu\text{m}$. This was nearly three times larger than the $D_{50} = 2.2 \mu\text{m}$ measurement for the RTAD particles. Also, it can be noted from Fig. 6 (b) that most of the particles produced by RTAD were in the range of 0.4–10 μm and >90 % of the particles were smaller than 5 μm in diameter. In contrast, spray-dried powder consists of particles ranging from 3–30 μm , with only approximately 25 % of particles being less than 5 μm in size. This fact makes the RTAD powder attractive for inhaled delivery applications, particularly because particles smaller than 5 μm in diameter can reach the deep parts of the lung, including the bronchioles and alveoli.³⁴ The span of the size distributions, computed as $\text{span} = (D_{90} - D_{10})/D_{50}$ (where D_{10}

and D_{90} represent the 10th and 90th percentiles, signifying that 10 % and 90 % of the sample particles are smaller than these respective values), also exhibited variations. The span for the RTAD powder, $\text{span} = 2.0$, was slightly larger than that of the spray-dried particles, $\text{span} = 1.7$, which could be explained by different utilized powder collection methods (filter for RTAD versus cyclone for spray drying).

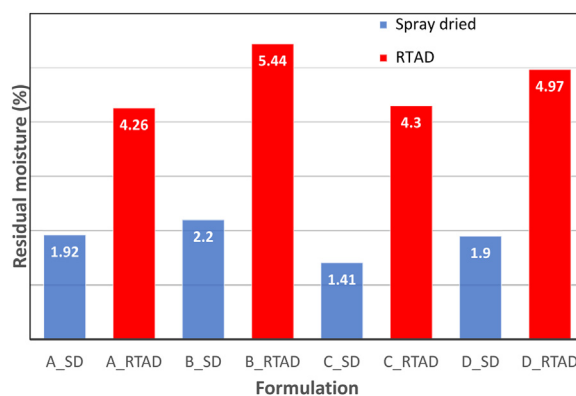


Figure 4. Residual moisture content for the two employed drying systems, and for different formulations. Here A–D labels denote the powders obtained by spray drying, A_RTAD–D_RTAD indicate the powders produced by the RTAD system.

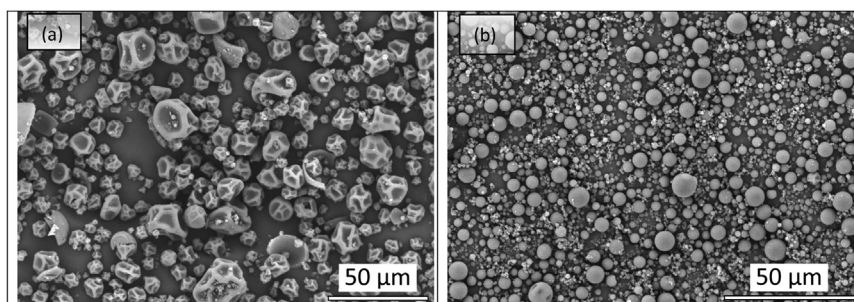


Figure 5. Typical SEM micrographs for (a) spray-dried powder and (b) powder produced by the RTAD setup.

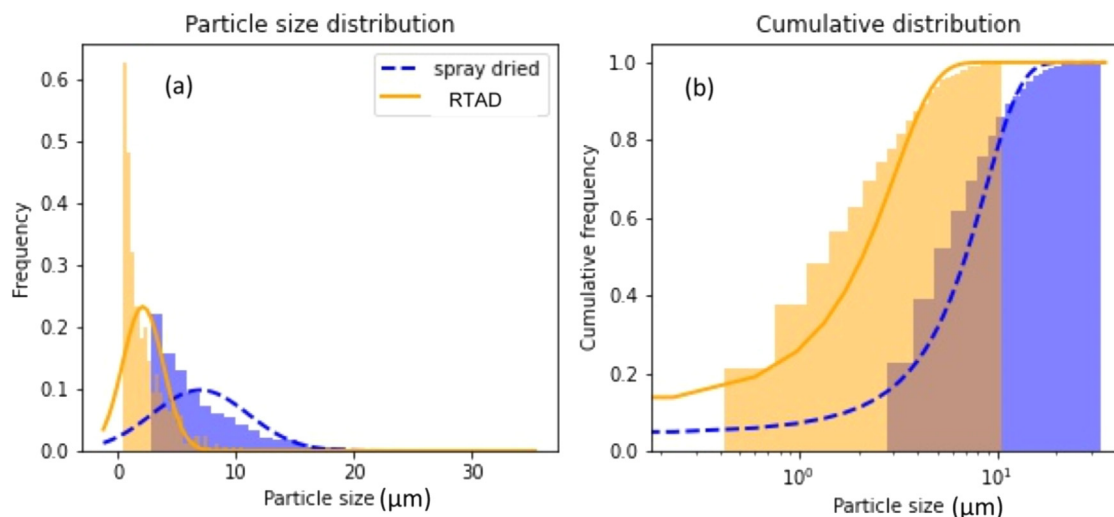


Figure 6. Number-weighted particle size distributions, obtained from analysis of SEM micrographs for spray-dried and RTAD powders. (a) Probability density function and (b) cumulative size distributions.

Finally, Table 3 summarizes the calculated yields, defined as the ratio of the recovered powder mass to the mass of the solid fraction initially supplied within the liquid solution that was sprayed into the drying chamber, for both processes across various formulations. In every case, spray drying consistently yielded higher results when compared to the RTAD method. This observed difference in yield can be primarily attributed to the distinct powder collection mechanisms employed by these two processes. The Buchi B-290 spray drying employs a cyclone separator, while RTAD utilizes a membrane filter. Additionally, differences in drying chamber designs, flow patterns, as well as the combined influence of smaller particle size and slightly higher dry particle moisture content in RTAD contribute to this discrepancy.

Notably, the analysis of product losses in the RTAD process revealed significant wall depositions within the conical region of the chamber, positioned just before the collecting filter (see Fig. 1).

Table 3
Yield for spray drying and Rapid Room-Temperature Aerosol Drying system (RTAD).

Sample	Spray dried (%)	RTAD (%)
A	86	53
B	64	44
C	72	34
D	88	46
Average	78	44

Consequently, there is a clear opportunity for enhancing the yield of the RTAD setup through future optimizations. Potential strategies for improvement include reducing the residual moisture content, fine-tuning the formulation, and enhancing the drying chamber design and gas-particle flow patterns through Computational Fluid Dynamics simulations.^{35,36} Furthermore, it's worth considering the substitution of the currently utilized membrane filter with a specialized cyclone separator, tailored for efficient collection of sub-10 μm particles with an efficacy of over 90 %. Such an approach has been successfully employed in related work (e.g., as exemplified in³⁷).

Conclusion

In light of the challenges faced by current manufacturing units for biopharmaceuticals, such as high physicochemical stresses, lengthy production times, excessive energy consumption, and sustainability concerns, there is a growing demand for gentler manufacturing processes that can maintain product quality while upholding continuous manufacturing and sustainability goals. In this paper we described the application of a novel dehydration system to produce biopharmaceuticals, building upon previous research at Princeton University. The system utilizes aerosol-assisted drying at room temperature. The study aims to evaluate the system's potential in preserving the quality of biopharmaceutical products during the dehydration process, comparing it to conventional spray drying methods.

Various tests were conducted to assess the protein quality of the synthesized powders. The results indicate that materials produced

using this new system exhibit reduced degradation compared to those obtained through traditional spray drying, as well as demonstrate spherical morphology of product, particles of solid-core microstructure, smaller size of particulates appropriate for inhalation delivery applications and improved reconstitution of dried powder. At the same time, the results highlight specific areas that require further improvements for future prototypes and potential commercial versions of the system. These areas include residual moisture content and yield, which are critical factors in the field of biopharmaceuticals. Overall, the innovative RTAD system holds promise for enhancing the efficiency and effectiveness of production methods in the biopharmaceutical industry. It has the potential to contribute to the development of high-quality biopharmaceutical products and may serve as an alternative to traditional drying technologies such as lyophilization and spray drying processes.

Data Availability

Some of the research data used in preparation of the manuscript is confidential and may not be available upon request. The rest of the data will be available upon request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Acknowledgments

The authors affiliated with AstraZeneca thank William (Tom) Leach, Ken Hinds and Marco E. Favretto for their invaluable comments and feedback on this manuscript. The authors affiliated with Princeton University extend their sincere appreciation to Dr. Mark Esposito (KayoThera), Dr. Kenric Hoegler (AstraZeneca), Kathleen Cantagallo (Drug Development Strategic Consulting), Dr. Frederick Simon Golec, Dr. Wenjie Wei (UPenn), Prabhpreet Gill (Princeton University) and Dr. Rina Dukor (BioTools) for their invaluable research insights and commentary on the obtained results. The authors affiliated with Princeton University acknowledge the support from the University City Science Center via QED Proof-of-Concept Program (grant number QED 2102) and the support from Princeton University via Intellectual Property Accelerator Fund program, and also acknowledge the use of Princeton's Imaging and Analysis Center, which is partially supported through the Princeton Center for Complex Materials (PCCM), a National Science Foundation (NSF)-MRSEC program (DMR-2011750).

Appendix

The code is developed in Python to extract size information from SEM images (Fig. A1). More information and a snippet of the code is available in <https://github.com/sadeghhdj/SEM-Particle-size-git>.

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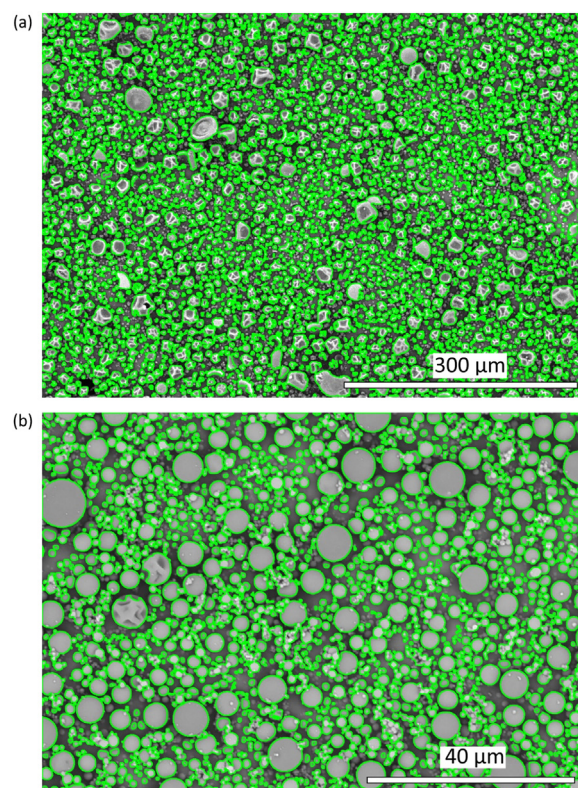


Figure A1. Image processing of the obtained SEM micrographs for (a) spray drying and (b) Rapid Room-Temperature Dehydration technologies. The developed computer tool automatically recognizes individual particles and marks them by green circles. The diameters and areas of green circles are further analyzed to construct the respective particle size distributions.

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